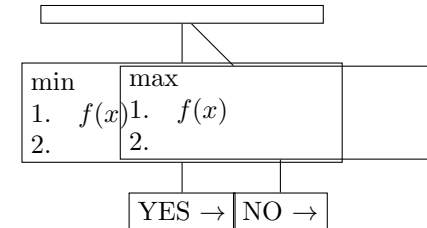


$$\begin{aligned} \min / \max \quad & f(x) & (1) \\ \text{s.t.} \quad & g_i(x) \leq b_i & (2) \\ & h_j(x) = c_j & (3) \\ & x \in X & (4) \end{aligned}$$

- **(Rank):** /
- : $C = AB$, $c_{ij} = \sum_k a_{ik} b_{kj}$
- : $AA^{-1} = I$
- $Ax = b$:
 - $m = n$, $\text{rank}(A) = n \rightarrow$
 - $m < n \rightarrow$
 - $\text{rank}(A) < \text{rank}([A|b]) \rightarrow$

- : $\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$
- : $\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_i \partial x_j} \end{bmatrix}$
- **1:** $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0)$
- **2:** $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2} (x - x_0)^T \nabla^2 f(x_0) (x - x_0)$

- **1:** $\nabla f(x^*) = 0$
- **2:**
 - min: $\nabla^2 f(x^*)$ (PD)
 - max: $\nabla^2 f(x^*)$ (ND)



(Linear Programming)

:

$$\begin{aligned} \min \quad & c^T x & (5) \\ \text{s.t.} \quad & Ax = b & (6) \\ & x \geq 0 & (7) \end{aligned}$$

:

- : n
- **(BFS):**
- : 0
- :
- :

- BFS**
- : ≤ 0
- :
- :
- :

(Transportation Problem)

$$\begin{aligned} \min \quad & \sum_i \sum_j c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_j x_{ij} \leq s_i \quad () \\ & \sum_i x_{ij} \geq d_j \quad () \\ & x_{ij} \geq 0 \end{aligned}$$

$$: \sum s_i < \max\{d_j\}$$

$$\begin{aligned} \min |x - a| \quad & \rightarrow \quad \min z \\ \text{s.t.} \quad & z \geq x - a \\ & z \geq -(x - a) \end{aligned}$$

$$\begin{aligned} X = \{x \in [a, b]^n : \sum x_i \geq c\} \\ \bullet : [a, b] \end{aligned}$$

- **LP**: Linear Programming ()
- **NLP**: Nonlinear Programming ()
- **BFS**: Basic Feasible Solution ()
- **PSD**: Positive Semidefinite ()
- **PD**: Positive Definite ()
- **s.t.**: subject to ()

- :
- :

$$\begin{aligned} (8) \quad & \bullet \ x^k \ (k \geq 1, \ x \geq 0): \\ (9) \quad & \bullet \ \ln(x) \ (x > 0): \\ & \bullet \ e^x: \\ & \bullet \ x^2 + y^2: \\ (10) \quad & \bullet \ \max\{f_1(x), f_2(x)\}: \ f_1, f_2 \\ (11) \quad & \bullet \ |x|: \end{aligned}$$

- ”here-and-now”:
- ”wait-and-see”:

$$\begin{aligned} (12) \quad & \\ (13) \quad & : \\ (14) \quad & \bullet : f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4} \text{ at } (3, 2) \end{aligned}$$

$$\nabla f = \left[\begin{array}{c} \frac{1}{2\sqrt{2}} \\ \frac{1}{4} \end{array} \right]$$

- : 2